Lie Algebra in Quantum Mechanics

Detailed Exposition

Contents

1 Introduction

Lie algebras form the mathematical backbone of symmetry in physics. In quantum mechanics, observables, generators of transformations, and conserved quantities are described using operator algebras that obey Lie algebra structures.

This document develops:

- Definition of Lie algebra
- Commutators and structure constants
- Lie algebras in quantum mechanics
- Hermitian operators and symmetry generators
- Examples: $\mathfrak{su}(2)$ and angular momentum
- Connection to Lie groups and unitary evolution

2 Definition of a Lie Algebra

A **Lie algebra** $\mathfrak{g}$ over a field (usually $\mathbb{C}$ or $\mathbb{R}$) is a vector space equipped with a bilinear operation called the **Lie bracket**

$$[\cdot, \cdot] : \mathfrak{g} \times \mathfrak{g} \to \mathfrak{g}$$

satisfying:

2.1 1. Bilinearity

$$[A + B, C] = [A, C] + [B, C]$$
$$[A, B + C] = [A, B] + [A, C]$$
$$[\lambda A, B] = \lambda[A, B]$$

2.2 2. Antisymmetry

$$[A, B] = -[B, A]$$

2.3 3. Jacobi Identity

$$[A, [B, C]] + [B, [C, A]] + [C, [A, B]] = 0$$

The Jacobi identity ensures internal consistency of nested commutators.

3 Lie Algebra in Quantum Mechanics

In quantum mechanics:

$$[A, B] := AB - BA$$

is the **commutator**.

The set of linear operators acting on a Hilbert space $\mathcal{H}$ forms a Lie algebra under the commutator.

3.1 Vector Space Structure

Operators satisfy:

$$(\lambda A + \mu B) |\psi\rangle = \lambda A |\psi\rangle + \mu B |\psi\rangle$$

Thus they form a complex linear vector space.

4 Hermitian Operators and Observables

In quantum mechanics, observables correspond to **Hermitian operators**:

$$A^\dagger = A$$

Properties:

- Eigenvalues are real
- Eigenvectors form an orthonormal basis
- Expectation values are real:
$$\langle\psi|A|\psi\rangle \in \mathbb{R}$$

If A and B are Hermitian:

$$[A, B]^\dagger = -[A, B]$$

Thus commutators of Hermitian operators are anti-Hermitian.

5 Generators of Continuous Transformations

If $U(\epsilon)$ is a continuous unitary transformation:

$$U(\epsilon) = e^{-i\epsilon G}$$

then G is Hermitian and called the **generator**.

Example: Time evolution

$$U(t) = e^{-iHt/\hbar}$$

where H is the Hamiltonian.
The generators form a Lie algebra because:

$$[G_i, G_j] = if_{ijk}G_k$$

where f_{ijk} are called **structure constants**.

6 Structure Constants

A Lie algebra can be written abstractly as:

$$[T_a, T_b] = f_{ab}{}^{c}T_c$$

The constants $f_{ab}{}^{c}$ define the algebra.
They must satisfy constraints from the Jacobi identity.

7 Example: The Lie Algebra $\mathfrak{su}(2)$

The most important example in quantum mechanics is angular momentum.
The generators J_x, J_y, J_z satisfy:

$$[J_x, J_y] = i\hbar J_z$$
$$[J_y, J_z] = i\hbar J_x$$
$$[J_z, J_x] = i\hbar J_y$$

This is the Lie algebra $\mathfrak{su}(2)$.
In compact notation:

$$[J_i, J_j] = i\hbar\epsilon_{ijk}J_k$$

where ϵ_{ijk} is the Levi-Civita symbol.

7.1 Physical Meaning

- Rotations do not commute.
- Successive infinitesimal rotations generate a new rotation.
- The noncommutativity encodes geometric curvature of rotation space.

8 Casimir Operator

A **Casimir operator** commutes with all generators:

$$[J^2, J_i] = 0$$

where

$$J^2 = J_x^2 + J_y^2 + J_z^2$$

This labels irreducible representations.
Eigenvalues:

$$J^2 |j, m\rangle = \hbar^2 j(j+1) |j, m\rangle$$

9 Lie Algebra vs Lie Group

9.1 Lie Group

A Lie group is a continuous group like:

$$SU(2), \quad SO(3), \quad U(1)$$

9.2 Lie Algebra

The Lie algebra is the tangent space at identity:

$$\text{Lie Group} \leftrightarrow \text{Exponentiation of Lie Algebra}$$

$$U = e^{-i\theta^a T_a}$$

Thus:

- Lie algebra = infinitesimal structure
- Lie group = finite transformations

10 Heisenberg Lie Algebra

Another important example:

$$[x, p] = i\hbar$$

This defines the Heisenberg algebra.
It encodes the uncertainty principle:

$$\Delta x \Delta p \geq \frac{\hbar}{2}$$

11 Anti-Commutator

Defined as:

$$\{A, B\} = AB + BA$$

Used in:

- Fermionic algebras
- Clifford algebra
- Supersymmetry

12 Representation Theory Insight

A representation assigns matrices to generators:

$$T_a \mapsto \rho(T_a)$$

such that:

$$\rho([T_a, T_b]) = [\rho(T_a), \rho(T_b)]$$

In quantum mechanics, representations determine:

- Spin values
- Particle multiplets
- Degeneracy structure

13 Deep Physical Insight

Lie algebras encode:

- Symmetry
- Conservation laws (Noether's theorem)
- Selection rules
- Degeneracies
- Quantum numbers

If:

$$[H, G] = 0$$

then G is conserved.

Thus Lie algebra structure directly governs dynamics and invariants of the system.

14 Summary

- Lie algebra = vector space + commutator
- Observables form Lie algebras
- Structure constants define symmetry
- $\mathfrak{su}(2)$ describes spin and angular momentum
- Heisenberg algebra describes position-momentum
- Exponentiation gives unitary evolution

Further Study

For deeper understanding:

- Representation theory
- Cartan subalgebra
- Root systems
- Semisimple Lie algebras
- Quantum groups